Esophageal Cancer Treatment

Esophageal Cancer Treatment Categories

Category	AJCC	Stage	Treatment
High-grade Dyplasia	0	Tis	Radiofrequency Ablation
Superficial Tumors	I	T1a	Endoscopic Therapy
Localized Tumors	IIB	T1b T2	Surgery
Locally-advanced	III IIA	T3 or N^+	Chemo $\pm$ RT $\rightarrow$ Surgery
Extra-Regional	IVA	N3	Chemo$\rightarrow$ ChemoRT$\rightarrow$ Surgery
Metastatic	IVB	M1	Chemotherapy $\pm$Radiation

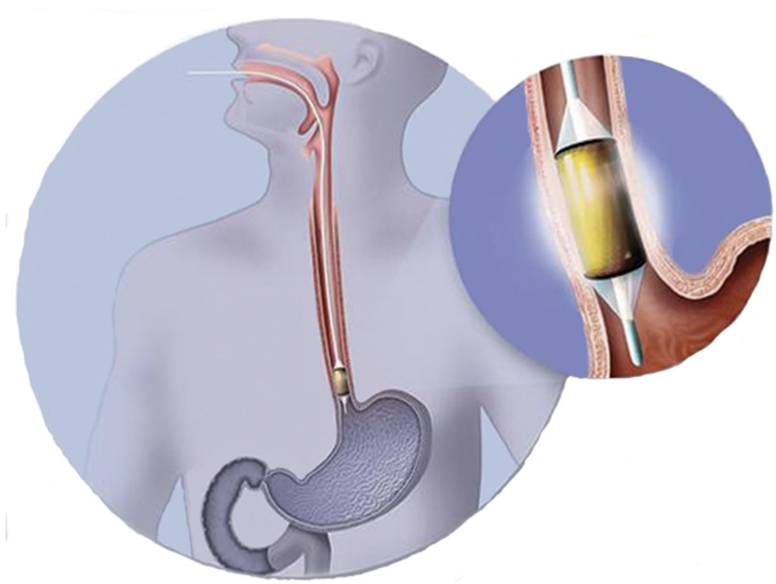
High-grade Dyplasia => Radiofrequency Ablation

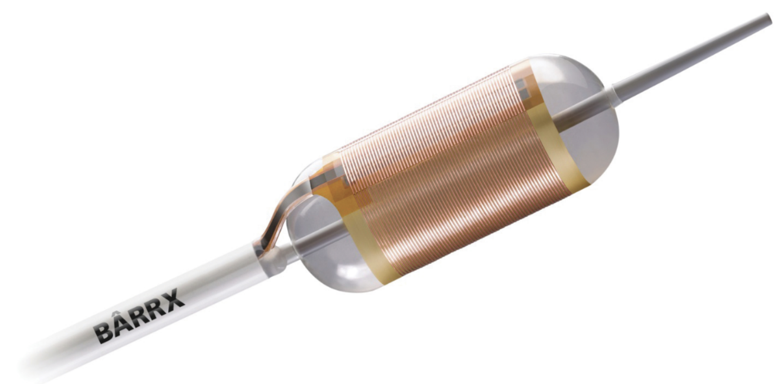
127 patients with dysplasia randomized:

- Radio-frequency ablation
- Sham ablation

Low-grade dysplasia in 64

High-grade dysplasia in 63

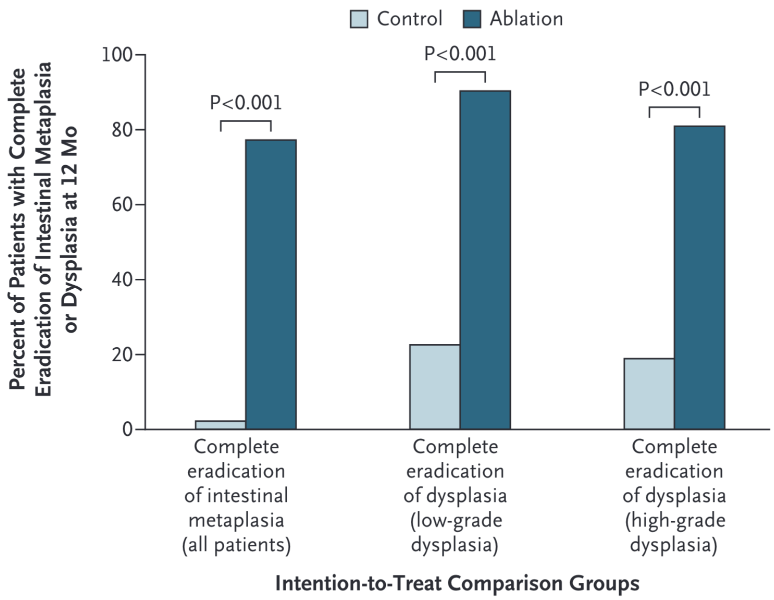




(Shaheen et al. 2009)

Radiofrequency Ablation for Dysplasia

Radiofrequency Ablation results in eradication of Barrett's in 75% at 1 year



No role for primary surgery in high-grade dysplasia

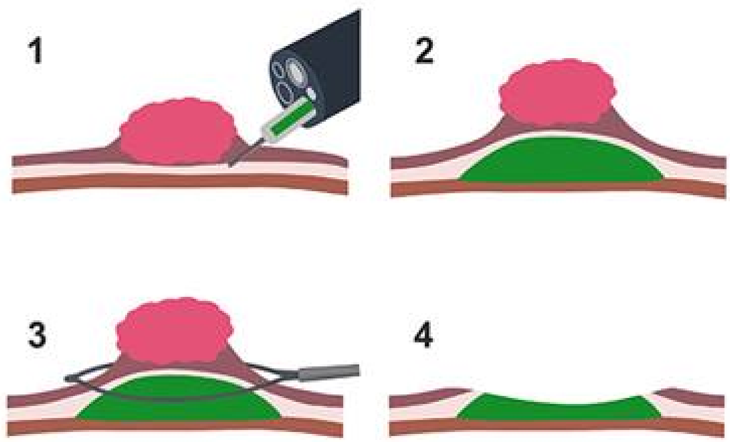
(Shaheen et al. 2009)

Superficial Tumors (T1)

Workup of nodular Barretts:

- Endoscopic Ultrasound
- Endoscopic Mucosal Resection
    - Diagnostic (T staging)
    - May be therapeutic for T1a tumors

Endoscopic Musocal Resection for T1 Tumors



Endoscopic Submucosal Dissection for T1a/T1b

Endoscopic resection uses needle-knife to dissect *below* submucosa

Maybe suitable for T1b lesions *if* lesion is completely resected

Risk of perforation

Localized Tumors (T2 N0)

Patients staged as uT2 N0 are candidates for primary surgery. *However:*

- EUS has a 25% rate of understaging uT2 N0 tumors
- Understaged patients who undergo primary surgery would need chemo or chemoRT postop

Asymptomatic Tumors (minimal dysphagia)

- EUS to distinguish T2 from T3 tumors
- If uT2 N0 → CT chest/abdomen/pelvis → Esophagectomy
- If uT3 or N1 → PET → neoadjuvant therapy

Patients with dysphagia almost always are T3 tumors (and don't need EUS)

Symptomatic Tumors (dysphagia)

Patients with dysphagia to solids or weight loss or tumor length >3cm are unlikely to have T1-2 tumors and can be initially evaluated with PET Scan

- Disease confined to the esophagus and regional nodes → Locally-advanced
- Metastatic disease → Metastatic
- N3 → Extra-Regional

EUS in Patients with Dysphagia

Memorial Sloan Kettering patients with esophageal cancer:

- 61 with dysphagia, 54 (89%) were found on EUS to have uT3-4 tumors.
- 53 without dysphagia, 25 (47%) were uT1-2 → candidates for primary surgery.

EUS can be omitted for patients with dysphagia

EUS can useful in patients *without* dysphagia (to identify T2N0 → Surgery)

(Ripley et al. 2016)

PET Scan

PET has more specificity and sensitivity than CT in detecting regional lymph node and distal metastasis

Locally-advanced

Improved survival with adjunctive therapy. There are two options:

- ChemoRT → Surgery (CROSS Trial)
- Chemo → Surgery → Chemo (EsoPEC Trial)

CROSS Surgery vs ChemoRT →Surgery

368 esophageal cancer patients randomized:

- Surgery alone
- Chemo+RT → Surgery

75% adenocarcinoma. T3: 80% T2: 17%. age $\tilde{x}$=60

⇒ longer survival with Chemo+RT → Surgery

Chemotherapy: Weekly carboplatin and paclitaxel

Radiation: 4140 cGy in 23 fractions (180cGy/fraction)

(Shapiro et al. 2015)

CROSS Surgery vs ChemoRT →Surgery

	Surgeruy	CRT→Surgery	
Time to Surgery	24d	97d	
Unresectable	13%	4%	$p=0.002$
R0 Resection	92%	69%	$p<0.001$
Positive Nodes	75%	31%	
Complications	∼	∼	
Mortality	∼	∼	

CROSS - Overall Survival

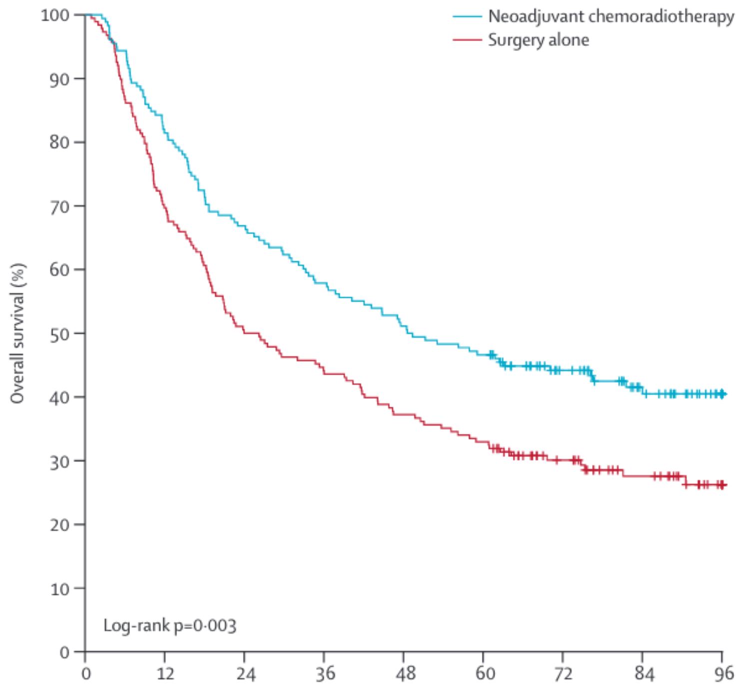


Figure 1: Surgery vs ChemoRT → Surgery

(Shapiro et al. 2015)

CROSS - Survival by Histology

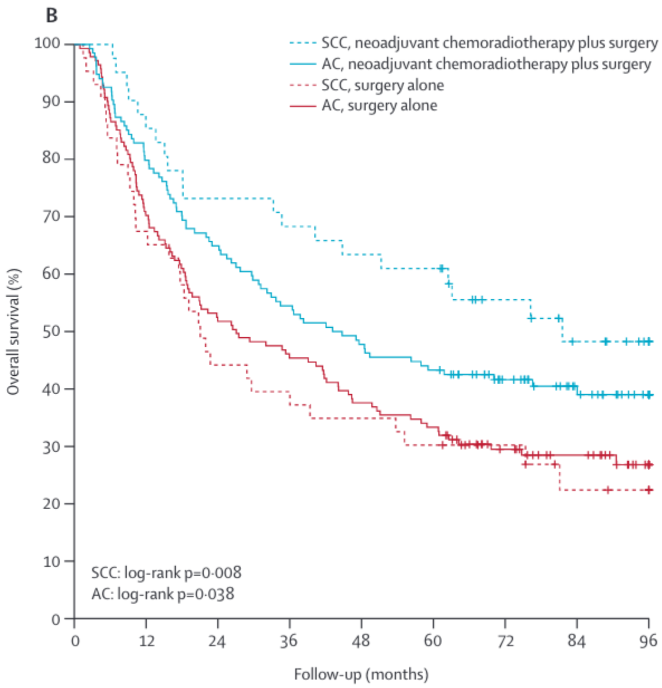


Figure 2: Surgery vs ChemoRT → Surgery

CROSS Effects by Histology

(Shapiro et al. 2015)

Adenocarcinoma

- Median survival 43mo vs 27mo
- Pathologic complete response in 23%

Squmous Cell Carcinoma

- Median survival 82mo vs 21mo
- Pathologic complete response in 49%

Surgical Therapy of Squamous Cell

(Shapiro et al. 2015)

Adjuvant Nivolumab after CROSS: Checkmate 577 Trial

EsoCA patients who received ChemoRT→ Surgery with residual disease (not pCR)

Randomized to one year of nivolumab vs Observation

⇒ Adjuvant nivolumab group had longer median survival: 22mo vs 11mo

(Kelly et al. 2021)

Checkmate 577 Trial

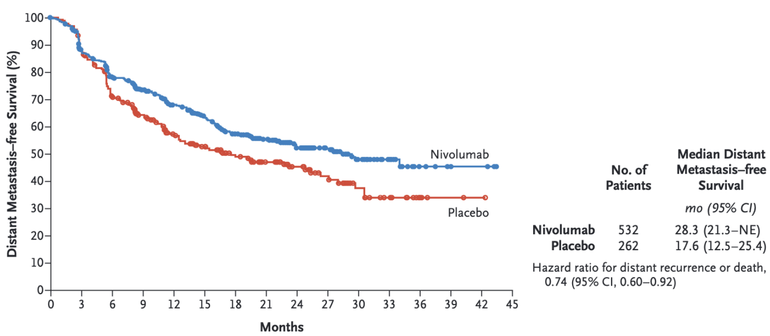


Figure 3: Adjuvant Nivolumab vs Observation

Nivolumab = Immune Checkpoint Inhibitor (ICI)

(Kelly et al. 2021)

PD-L1 agonist ligand

Interferes with tumor cell down-regulation of T cells

Active against stage IV esophageal cancer

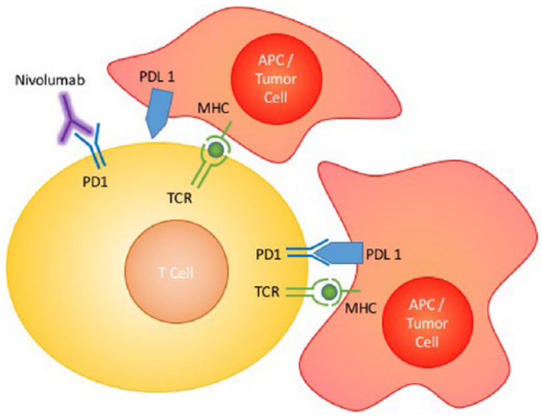


Figure 4: Nivolumab mechanism of action

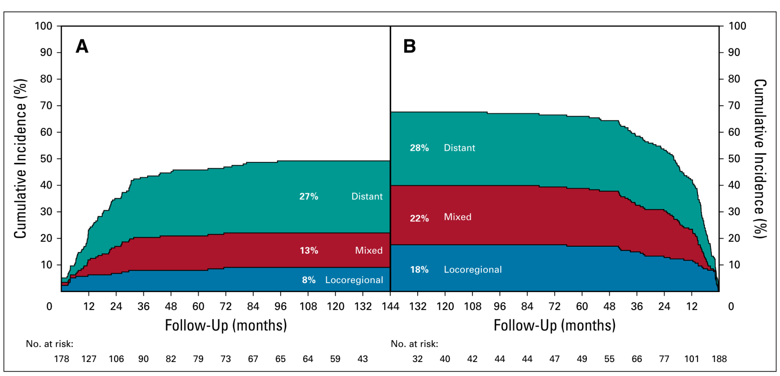
CROSS - ChemoRT Does Not Reduce Distant Failure

Sites of failure over time

ChemoRT + Surgery *vs* Surgery

ChemoRT appears to reduce risk of local or local+distant failure, but not isolated distant failure

⇒ Can chemotherapy reduce risk of distant failure?



(Shapiro et al. 2015)

EsoPEC CROSS vs FLOT for Adenocarcionoma

Adenocarcinoma esophagus and GE junction

- T1 N+ or T2-4a M0. Median age =63. 89% men

Randomized to CROSS (n=217) vs FLOT chemotherapy (n=221)

- CROSS: carboplatin/paclitaxel + 4140cGy → Surgery
- FLOT: FLOT → Surgery → FLOT

Excluded: Squamous cell, gastric cancer, T1N0, T4b, M1

(Hoeppner et al. 2025)

EsoPEC: FLOT superior to CROSS

	FLOT Periop Chemo	CROSS ChemoRT	CROSS Trial ChemoRT
90d Mortality	3.2%	5.6%	
$\tilde{x}$ Survival	66mo	37mo	43mo
3year Survival	57%	51%	58%
5year Survival	51%	29%	47%
path CR	17%	10%	23%

See also Neoadjuvant Chemo

(Hoeppner et al. 2025)

Surgery for Squamous Cell Carcinoma

Squamous Cell Carcinoma of the esophagus

- responds well to chemo+RT
- more difficult to get a surgical margin on the airway
- additional benefit of surgery on top of chemoRT is uncertain

Squamous Cell: ChemoRT $\pm$ Surgery (FFCD)

Squamous cell of the esophagus $\rightarrow$ 4500cGy RT + 2 cycles of cisplatin + 5FU

Patients with a clinical response were randomized:

- Surgery $\Rightarrow$ 2 year survival 34% Median 17.7mo
- 3 cycles of chemo + 2000 cGy RT $\Rightarrow$ 2 year survival 40% Median 19.3mo

$\Rrightarrow$ No difference in overall survival

(Bedenne et al. 2007)

Squamous Cell: ChemoRT $\pm$ Surgery (German)

Squamous cell of the esophagus randomized:

- 4000 cGY RT + Chemo $\rightarrow$ Surgery. 64% 2-year PFS. Mortality 12.8%
- 6500cGy RT + Chemo: 41% 2-year PFS. Mortality 3.5%

$\Rrightarrow$ No difference in overall survival

(Stahl et al. 2005)

Non-Operative Management

Extra-regional Disease = N3

Patients with N3 disease are Stage IVA and have a poor prognosis.

Treatment consists of chemotherapy followed by radiation.

Patients with persistent disease in the esophagus (and no distant disease) would then be candidates for surgery

Metastatic = IVB

FOLFOX is first-line systemic therapy for metastatic GI cancers

- Dose-limiting toxicity is frequently peripheral neuropathy

Adjuvant Immunotherapy: Checkmate 577 Trial

Immunotherapy with nivolumab as adjuvant therapy after CROSS regimen for patients with residual disease

Stage II/II Esophageal or GE junction cancers Adenocarcinoma or squamous cell

ChemoRT → Surgery *with residual disease on pathology*

Treatment Group: Nivolumab every 2 weeks x 4 months → every month x 8 months

vs

Control Group: No adjuvant therapy

⇒ Better survival in group with adjuvant nivolumab

(Kelly et al. 2021)

Neoadjuvant Chemo for EsoCA

- MAGIC trial (gastric): ECF[1]→Surgery→ECF *vs* Surgery
- OEO2 Trial: (esophageal) Chemo→Surgery→ Chemo *vs* Surgery
- FLOT (gastric): FLOT[2]→Surgery→ FLOT *vs* ECF→Surgery→ECF
- EsoPEC: (esophageal):FLOT→Surgery→FLOT *vs* ChemoRT→Surgery (CROSS)

[1]Epirubicin, Cisplatin, 5FU
[2]5FU, Leuvocorin, Oxaliplatin, Decetaxol

OEO2 Clinical Trial

- 802 Esophageal adenocarcinoma and squamous cell
- Randomized to Chemo → Surgery → Chemo *vs* Surgery alone
- Chemotherapy with ECF (Epirubicin, Cisplatin, 5FU)
- 5-year survival 23% for chemo+surgery vs 17% for surgery (HR 0.84 p=0.03)

(Allum et al. 2009)

Neo-Aegis Tral CROSS vs MAGIC/FLOT

- **Adenocarcinoma** T2-3 N0-3 M0 Tumor length <8cm
- ChemoRT arm: carboplatin + paclitaxel + 4140cGy
- Chemo arm: MAGIC (ECF) or FLOT (later in trial)
- No difference in overall survival
- R0 resection 96% with CROSS vs 82% with chemo
- pCR 12% with CROSS vs 4% with chemo

(**reynolds1015?**)

Orientation Manual

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